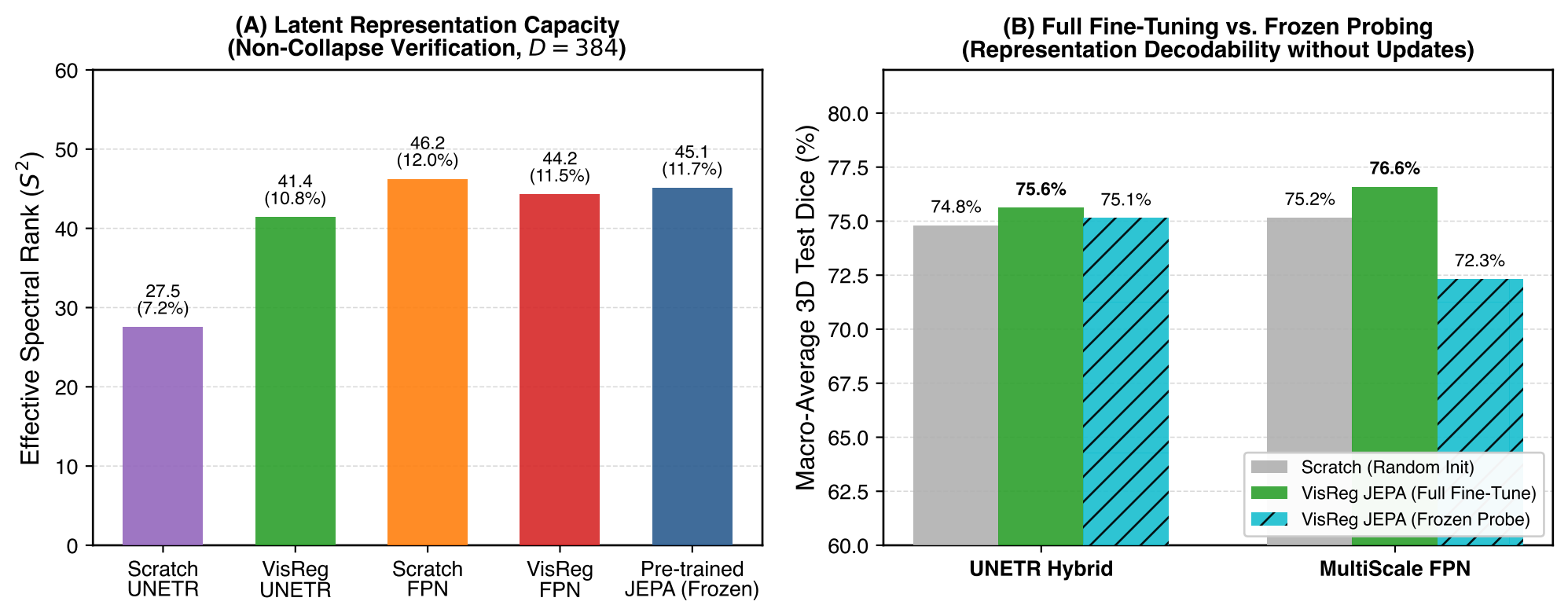
(A) Latent Representation Capacity
(Non-Collapse Verification, $D = 384$)
Effective Spectral Rank (S^2)
0
10
20
30
40
50
60
27.5
(7.2%)
41.4
(10.8%)
46.2
(12.0%)
44.2
(11.5%)
45.1
(11.7%)
Scratch
UNETR
VisReg
UNETR
Scratch
FPN
VisReg
FPN
Pre-trained
JEPA (Frozen)
(B) Full Fine-Tuning vs. Frozen Probing
(Representation Decodability without Updates)
Macro-Average 3D Test Dice (%)
60.0
62.5
65.0
67.5
70.0
72.5
75.0
77.5
80.0
74.8%
75.6%
75.1%
75.2%
76.6%
72.3%
UNETR Hybrid
MultiScale FPN
Scratch (Random Init)
VisReg JEPA (Full Fine-Tune)
VisReg JEPA (Frozen Probe)